Texture Mapping and Distortion Analysis

Project Outline, ECE 467/567 Computer Graphics and CAD, Fall 2026

Tanner Muscarella Drake Kendall

Revised October 27, 2026, after proposal feedback

1 Status

The proposal was approved with a score of 9/10, on the condition that the core is trimmed to a single stack (Python and Polyscope). This outline reflects that change and the instructor's suggestions: a cut step for closed meshes, pinned vertices for LSCM, theory-based tests, and two focused evaluations.

2 Goal

Build a Python tool that maps 2D images onto curved 3D surfaces using parameterization methods we implement ourselves, and that measures and visualizes the distortion each method unavoidably introduces. By Gauss's *Theorema Egregium*, no surface with nonzero Gaussian curvature can be flattened without distortion, so every method trades off angle preservation against area preservation. The tool makes that trade-off visible and measurable. As a stretch goal, the mapped images are projected onto real objects with a calibrated projector.

3 Core (Python and Polyscope only)

1. **Test meshes:** sphere, hemisphere, torus and disk-like patches, generated analytically or loaded from `.obj` files with trimesh.
2. **Analytic sphere projections:** equirectangular, Mercator and stereographic.
3. **Tutte embedding** with uniform or mean-value weights: fix the boundary loop to a circle and solve a sparse linear system for the interior vertices.
4. **Least Squares Conformal Maps (LSCM):** free boundary, with two boundary vertices pinned far apart to remove the null space (translation, rotation and scale) so the system is not singular.
5. **Cut step for closed meshes:** Tutte and LSCM require disk topology. A closed genus-0 mesh is cut along a shortest edge path (Dijkstra) between two chosen vertices to create one boundary loop; a torus needs two cuts (a meridian and a longitude). Methods are tested on disks first.
6. **Distortion analysis:** for each triangle, the Jacobian J of the map from the triangle's 3D plane to UV space, and its singular values $\sigma_1 \geq \sigma_2$:

$$\text{angle distortion} = \frac{\sigma_1}{\sigma_2}, \qquad \text{area distortion} = \sigma_1\sigma_2, \qquad \text{fold-over} \iff \det J < 0.$$

7. **Viewer (Polyscope):** switch between methods, view the 3D mesh and its 2D UV layout, and toggle distortion heatmaps and a checkerboard texture.

Built-in tests from theory

- Tutte with positive weights and a convex boundary must give zero fold-overs.
- Mercator is conformal, so measured σ_1/σ_2 should be close to 1 away from the poles.
- Unit tests on mesh generation: vertices on the sphere, valid face indices, no degenerate triangles, outward normals.

4 Evaluation

Two evaluations, done thoroughly:

1. **Distortion by method and mesh:** a table of mean, 95th-percentile and maximum angle and area distortion, plus fold-over counts, for each method on each test mesh, with heatmap figures.
2. **Solve time vs. mesh size:** timing for Tutte and LSCM across mesh resolutions, on a log-log plot.

libigl's Python bindings are used only to validate our implementations during development, not as part of the submitted code or as a reported result.

5 Stretch goals

Each can be dropped without affecting the core.

- **Cotangent (harmonic) weights:** these go negative when the two angles opposite an edge sum to more than 180°, so fold-overs are reported separately for this case.
- **ARAP parameterization** (Liu et al. 2008).
- **Multi-chart atlas** with chart segmentation.
- **Procedural 3D shaders** that need no UV map (moved from the core).
- **Phone-scanned objects** with mesh cleanup (moved from the core).
- **Web front end** (React, Three.js, GLSL), lowest priority.
- **Projection mapping ladder:**
  - A1: project onto a known shape (sphere or box), pose from clicked correspondences (`cv2.solvePnP`), rendering from the projector's viewpoint.
  - A2: project onto a phone-scanned object, registered manually and UV-mapped with our code.
  - A3: structured-light (Gray code) scanning with the projector and a camera, giving a mesh already in the projector's frame.

6 Tools

Purpose	Tool
Language	Python
Math and sparse solves	NumPy, SciPy
Mesh input	trimesh (file I/O only)
Viewer	Polyscope
Plots	Matplotlib
Tests	pytest
Validation only	libigl Python bindings
Stretch (projector)	OpenCV, projector, webcam, phone photogrammetry app
Collaboration	Git and GitHub

7 Paper presentations

Individual; each of us presents a different paper. Choices are due Wed Oct 28.

- Lévy, Petitjean, Ray and Maillot, *Least Squares Conformal Maps for Automatic Texture Atlas Generation*, SIGGRAPH 2002.
- Sander, Snyder, Gortler and Hoppe, *Texture Mapping Progressive Meshes*, SIGGRAPH 2001. Its stretch metric uses the same singular-value idea as our distortion maps.

8 Schedule

Date	Deliverable	Target
Wed Oct 28	Paper choices	LSCM and Sander et al., one each
Thu Nov 5	Paper presentations	Each paper plus related works
Tue Nov 17	Milestone	Viewer with sphere projections and Tutte, heatmaps, theory tests passing
Late Nov	Core complete	LSCM, cut step, both evaluations
Tue Dec 8	Demo and final report	Live demo, backup video, report and code

The task-level schedule with dependencies is in the separate task table document.

9 Risks

- **LSCM numerics:** a singular system if the pins are missing or poorly placed. Mitigated by pinning two distant boundary vertices and validating against libigl.
- **Cutting closed meshes:** seam paths can produce awkward boundaries. Mitigated by testing on disks first and keeping atlas segmentation as stretch.
- **Hardware access** for the projector stretch goals: borrow and test early.

References

- K. Hormann, B. Lévy, A. Sheffer, *Mesh Parameterization: Theory and Practice*, SIGGRAPH Course Notes, 2007.

- B. Lévy, S. Petitjean, N. Ray, J. Maillot, *Least Squares Conformal Maps for Automatic Texture Atlas Generation*, SIGGRAPH 2002.
- P. Sander, J. Snyder, S. Gortler, H. Hoppe, *Texture Mapping Progressive Meshes*, SIGGRAPH 2001.
- M. Floater, *Mean Value Coordinates*, CAGD, 2003.
- W. T. Tutte, *How to Draw a Graph*, Proc. London Mathematical Society, 1963.
- L. Liu, L. Zhang, Y. Xu, C. Gotsman, S. Gortler, *A Local/Global Approach to Mesh Parameterization*, SGP 2008.
- J. Snyder, *Map Projections: A Working Manual*, USGS Professional Paper 1395, 1987.
- R. Raskar, G. Welch, K.-L. Low, D. Bandyopadhyay, *Shader Lamps: Animating Real Objects with Image-Based Illumination*, EGWR 2001.